

[Dashboard](#) / [My courses](#) / [MA-224-G 24H](#) / [Tests](#)/ [Test 4 \(topics 10-12: Applications of Lagrange's theorem,Coding theory,Coding theory and group codes\)](#)**Started on** Wednesday, 20 November 2024, 2:26 PM**State** Finished**Completed on** Wednesday, 20 November 2024, 2:52 PM**Time taken** 26 mins 8 secs**Marks** 2.00/3.00**Grade** **2.00** out of 3.00 (**66.68%**)

Information

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This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade.**

The following rules apply:

- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question 1

Incorrect

Mark 0.00 out of 1.00

If p and q be prime numbers, let $n = p \cdot q$, and write (n, x) and (n, y) for the public and private key in an RSA key pair associated to n .

In the list below we have written suggestions for $(x, y, p, q, p \cdot q)$. Some of them give valid RSA keys, some of them do not for different reasons.

Select all options that give us a valid key pair. Leave the invalid options un-checked.

- ☒ $x = 17570399, y = 30362099, p = 7451, q = 6451, n = 48066401$
- ☐ $x = 12903033, y = 3215451, p = 6113, q = 2437, n = 14897381$
- ☒ $x = 20271383, y = 22663367, p = 3557, q = 7741, n = 27534737$
- ☒ $x = 2846781, y = 2055541, p = 2709, q = 2861, n = 7750449$
- ☐ $x = 7057729, y = 1240783, p = 6007, q = 3271, n = 19648897$
- ☒ $x = 42185665, y = 31377985, p = 7249, q = 6329, n = 45878921$
- ☐ None of the above

You are free to use this calculator for products $\bmod (n)$. For example, if you want to compute $(7634 \cdot 99833) \% 8883993$, you should set **Factor 1**=7634, **Factor 2**=99382, and **Modulus**=8883993.

Factor 1:
factor 2:
Modulus:
Result:

You are free to use this calculator for computing powers $\bmod (n)$. For example, if you want to compute $(7634^{99382}) \% 8883993$, you should set **Base**=7634, **Exponent**=99382, and **Modulus**=8883993.

Base:
Exponent:
Modulus:
Result:

Question **2**

Correct

Mark 1.00 out of 1.00

Compute the minimal and maximal Hamming distance between any two bitstrings in the set

[001011, 110000, 000010, 000111]

.

$l_{min} =$

Your last answer was interpreted as follows:

2

$l_{max} =$

Your last answer was interpreted as follows:

5

Question **3**

Correct

Mark 1.00 out of 1.00

Let

$$E : (\mathbb{Z}/2)^{\times 3} \rightarrow (\mathbb{Z}/2)^{\times 5}$$

be the encoding function given by $E(w_1 w_2 w_3) = w_1 w_2 w_3 w_4 w_5$, where

$$\begin{bmatrix} w_4 = w_2 \\ w_5 = 0 \end{bmatrix}$$

a) When sending a codeword x , what is the maximum number of transmission bit errors this encoding is always able to detect?

Your last answer was interpreted as follows:

0

b) Find a **non-zero** codeword c for the encoding E .

$c =$

Your last answer was interpreted as follows:

00100

Any codeword will do, as long as it is not the bitstring consisting only of zeros **000...0**.

c) Find a bitstring x in $(\mathbb{Z}/2)^{\times 5}$ which is **not** a codeword for the encoding E .

$x =$

Your last answer was interpreted as follows:

11111

Syntax guide: Give your answers to **b)** and **c)** as bitstrings. E.g. if you think the answer to a question is "1011", then you should enter **1011**.

Question **4**

Correct

Mark 0.00 out of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: 212

Signing code:

Your last answer was interpreted as follows:

46400

◀ [Test 3 \(topics 7-9: Recurrence relations, Introduction to group theory, The multiplicative group of integers mod\(n\) and Lagrange's theorem\)](#)

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